

1.INTRODUCTION:

Automation is described as a technology that operates by accepting commands from a programmed code that has feedback control built in. As a result, this kind of system control the loads with less human interaction. Home automation has become an essential part of modern smart living, offering convenience and enhanced control over household appliances. With the growth of Internet of Things (IoT) platforms, implementing smart home systems has become more accessible, one such powerful platform is **Blynk IoT**, which provides a user-friendly interface to monitor and control devices remotely through a smartphone or web dashboard. In this project, a home automation system is developed using microcontroller ESP32 interfaced with the Blynk IoT platform. The system allows users to control lights, fans, and other appliances from anywhere using the internet.

1.1 OBJECTIVES:

- ❖ To design a basic home automation system that allows control of household appliances through the Blynk IoT platform.
- ❖ To enable users to operate appliances remotely using a smartphone and an internet connection.
- ❖ To establish communication between the microcontroller (ESP32) and the Blynk cloud for reliable device control.
- ❖ To enhance user convenience and comfort by reducing the need for manual operation of appliances.
- ❖ To improve energy usage awareness by allowing users to check the status of devices and avoid leaving them ON unnecessarily.

1.2 METHODOLOGY:

❖ Problem Identification and Requirement Analysis

The project begins by identifying the need for a system that allows users to control household appliances remotely. The required components such as the ESP8266/ESP32 microcontroller, relay modules, power supply, and the Blynk IoT application are selected based on their availability and suitability.

❖ Hardware Setup

- i. The microcontroller (ESP32) is used as the central control unit.
- ii. Appliances are connected through relay modules.
- iii. The relay acts as a switch that the microcontroller controls electronically.
- iv. Proper wiring is done to ensure safe and reliable switching of AC appliances.

❖ Software Configuration

- i. The Blynk IoT platform is used to create a mobile dashboard.
- ii. A new project is created in the Blynk app.
- iii. Virtual buttons and indicators are added to control and monitor appliances.
- iv. An authentication token is generated to link the app with the microcontroller.

❖ Microcontroller Programming

- i. The microcontroller is programmed using the Arduino IDE.
- ii. The Wi-Fi credentials and Blynk authentication token are included in the code.
- iii. Each device is assigned to a virtual pin in Blynk.
- iv. The code enables communication between the microcontroller and the Blynk cloud so that relay states can be updated based on user inputs.

❖ **Establishing IoT Connectivity**

- i. The system is connected to a Wi-Fi network.
- ii. Once powered, the microcontroller connects to the internet and logs into the Blynk cloud.
- iii. This connection allows real-time control and status feedback.

❖ **Integration and Testing**

- i. After completing the hardware and software setup:
- ii. Each appliance is tested using manual switches and virtual buttons in the Blynk app.
- iii. ON/OFF commands are verified to ensure proper relay switching.
- iv. The status of each device is monitored to confirm reliable feedback.

1.3 COMPONENTS:

- **ESP32:** It is a microcontroller with in built WiFi and Bluetooth, Controls the relays ON/OFF condition by providing the electrical signal to the relays.
- **Relays:** The 4-channel relay module is an electrically operated switch used to control high-voltage or high-current loads using low-voltage signals from the ESP32. Each relay can independently turn ON or OFF the appliances connected to it. The module operates on 5V dc and can drive loads up to 10A, making it suitable for controlling the loads. In this project, the relay module acts as an interface between the low-power ESP32 and the high-power loads.
- **Push buttons:** Push buttons are simple input devices that make or break a connection when pressed. These buttons allow manual control of the loads even when Wi-Fi or the Blynk app is unavailable. Each button corresponds to one load and acts as an additional way to operate the home automation system locally.

2.CIRCUIT DIAGRAM:

Circuit diagram include ESP32, 4channel relay module(10A), push buttons and different types of loads like synchronous motor, LED lamp, incandescent lamp and socket. The connections of all these components are made as shown in the figure1.

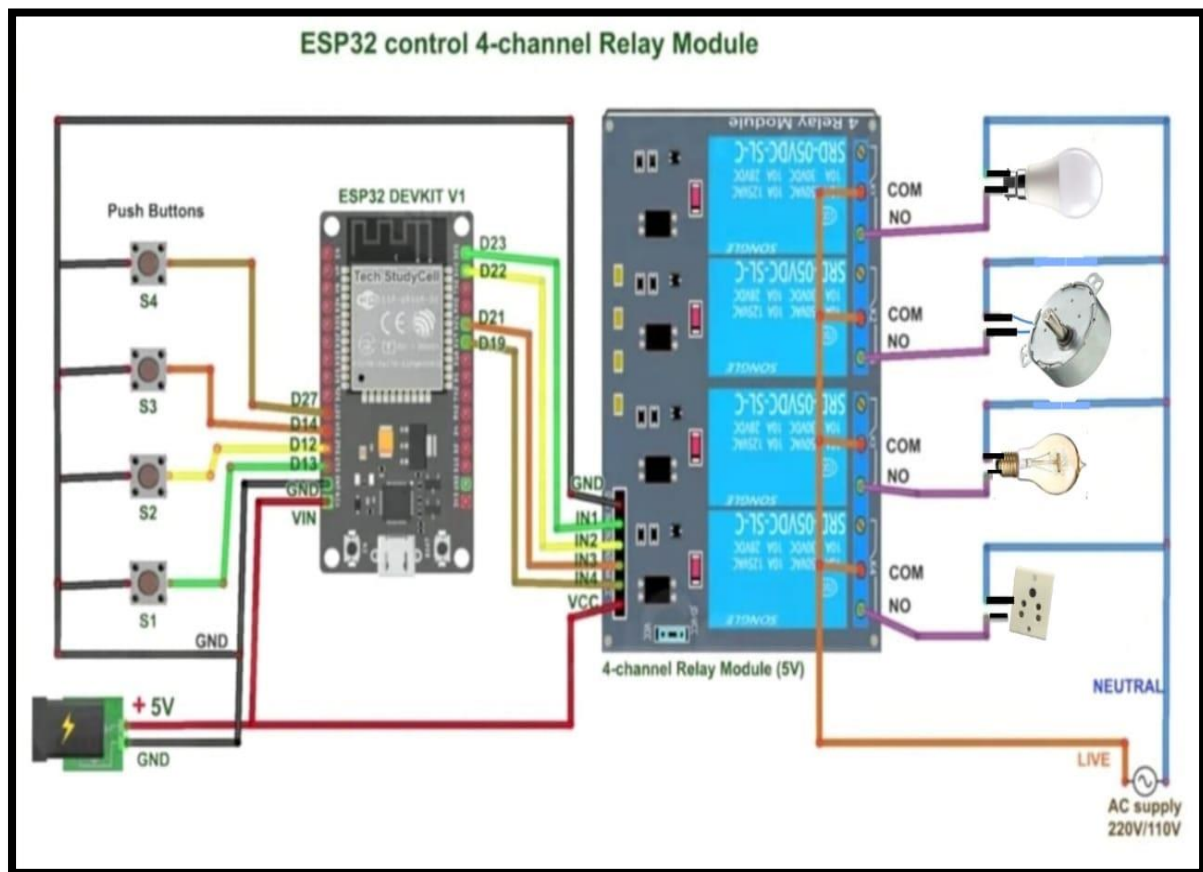


Figure 1: Circuit Diagram

3.FLOWCHART:

Flowchart shows two ways of controlling load status by Blynk application and by manual switches with Wi-Fi and without Wi-Fi as shown in the figure2.

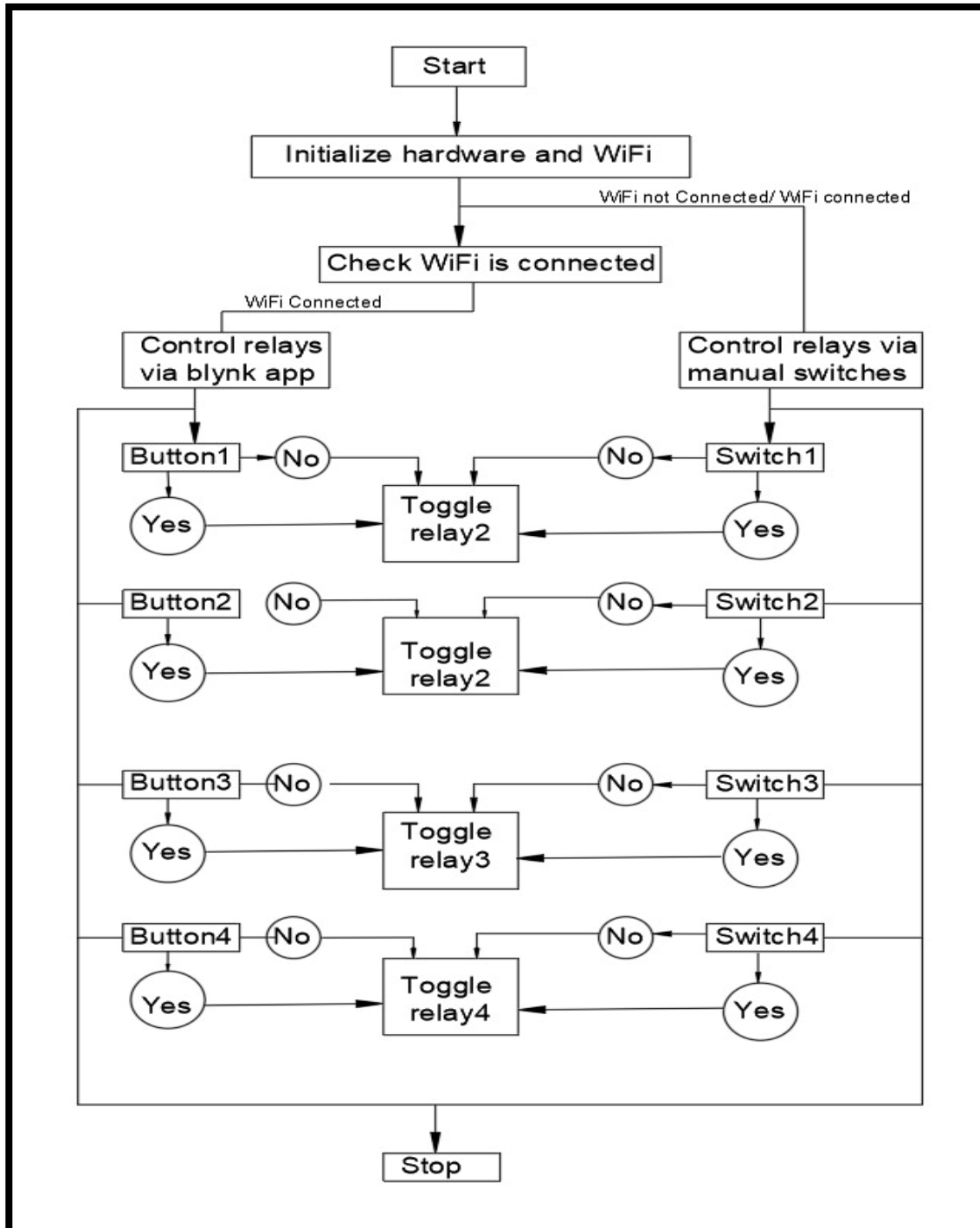


Figure 2: flowchart

4.ADVANTAGES:

- ❖ Home automation is the cost effective method.
- ❖ It is safe then the conventional method used in the homes.
- ❖ We can control the loads from anywhere.
- ❖ There are two ways to control the load by manual switches and through mobile.

5.RESULT:

Once testing is successful, the setup is arranged neatly for demonstration. The system can now be control through mobile using Blynk IOT and by manual switches, providing convenience and safety awareness. By using Blynk and manual switches status of the following load is controlled.

Blynk Switch	Manual Switch	Load
Button 1	Switch1	Motor
Button 2	Switch 4	Color LED
Button 3	Switch 3	LED
Button 4	Switch 2	Extension Board

6.Conclusion:

The home automation system developed in this project successfully demonstrates how household appliances can be controlled remotely and efficiently using the ESP32 microcontroller, relay module, and the Blynk IoT platform. The system allows users to operate the loads through the mobile application and manual push buttons, ensuring flexibility and reliability. The integration of Wi-Fi communication enables real-time monitoring and control from anywhere, improving convenience and safety. Overall, the project shows that IoT-based automation is cost-effective, easy to implement and can be expanded to control more devices, making it a practical solution for modern smart homes.

7.Reference:

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